

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.

5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.

4. Identify the **default gateway** that hosts in each LAN should use for communication.

Solutions:

QUESTION 1: Smart Campus Network Subnetting

Given:

- Network Address = 192.168.100.0/24
- School of Computing = /25
- School of Electronics = /26
- School of Mechanical Engineering = /27

Step 1: School of Computing (/25)

Subnet Mask = 255.255.255.128

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 25)} - 2 \\ &= 128 - 2 \\ &= 126 \text{ Hosts}\end{aligned}$$

Network Address : 192.168.100.0

First Host Address : 192.168.100.1

Last Host Address : 192.168.100.126

Broadcast Address : 192.168.100.127

Usable Hosts : 126

Step 2: School of Electronics (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2\end{aligned}$$

= 62 Hosts

Network Address : 192.168.100.128

First Host Address : 192.168.100.129

Last Host Address : 192.168.100.190

Broadcast Address : 192.168.100.191

Usable Hosts : 62

Step 3: School of Mechanical Engineering (/27)

Subnet Mask = 255.255.255.224

Formula:

Usable Hosts = $2^{(32 - 27)} - 2$

= $32 - 2$

= 30 Hosts

Network Address : 192.168.100.192

First Host Address : 192.168.100.193

Last Host Address : 192.168.100.222

Broadcast Address : 192.168.100.223

Usable Hosts : 30

Step 4: Remaining Unused IP Address Range

Remaining Address Range

Start Address : 192.168.100.224

End Address : 192.168.100.255

Final Result Table

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Remaining Unused IP Address Range

Start Address	End Address
192.168.100.224	192.168.100.255

Interpretation of Results

The Smart Campus network 192.168.100.0/24 is successfully divided into three subnets according to the departmental requirements.

- School of Computing (/25) supports 126 usable host addresses.
- School of Electronics (/26) supports 62 usable host addresses.
- School of Mechanical Engineering (/27) supports 30 usable host addresses.

The remaining IP address range (192.168.100.224 – 192.168.100.255) is reserved for future network expansion. This subnetting design improves efficient IP address utilization, network performance, security, and simplifies network administration.

QUESTION 2: Multinational Software Company Network Subnetting

Given:

- Network Address = 10.10.0.0/16
- Software Development = /24
- Quality Assurance = /25
- Human Resources = /26
- Executive Management = /27

Step 1: Software Development (/24)

Subnet Mask = 255.255.255.0

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 24)} - 2 \\ &= 256 - 2 \\ &= 254 \text{ Hosts}\end{aligned}$$

Network Address : 10.10.0.0
First Host Address : 10.10.0.1
Last Host Address : 10.10.0.254
Broadcast Address : 10.10.0.255
Usable Hosts : 254

Step 2: Quality Assurance (/25)

Subnet Mask = 255.255.255.128

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 25)} - 2 \\ &= 128 - 2 \\ &= 126 \text{ Hosts}\end{aligned}$$

Network Address : 10.10.1.0
First Host Address : 10.10.1.1
Last Host Address : 10.10.1.126
Broadcast Address : 10.10.1.127
Usable Hosts : 126

Step 3: Human Resources (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2 \\ &= 62 \text{ Hosts}\end{aligned}$$

Network Address : 10.10.1.128
 First Host Address : 10.10.1.129
 Last Host Address : 10.10.1.190
 Broadcast Address : 10.10.1.191
 Usable Hosts : 62

Step 4: Executive Management (/27)
 Subnet Mask = 255.255.255.224

Formula:

$$\begin{aligned}
 \text{Usable Hosts} &= 2^{(32 - 27)} - 2 \\
 &= 2^5 - 2 \\
 &= 30 \text{ Hosts}
 \end{aligned}$$

Network Address : 10.10.1.192
 First Host Address : 10.10.1.193
 Last Host Address : 10.10.1.222
 Broadcast Address : 10.10.1.223
 Usable Hosts : 30

Step 5: Remaining Unused IP Address Range
 Remaining Address Range

Start Address : 10.10.1.224
 End Address : 10.10.255.255

Final Result Table

Department	Subnet Mask	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24 (255.255.255.0)	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25 (255.255.255.128)	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26 (255.255.255.192)	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27 (255.255.255.224)	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Remaining Unused IP Address Range

Start Address	End Address
10.10.1.224	10.10.255.255

Interpretation of Results

The IP block 10.10.0.0/16 has been successfully subnetted to meet the requirements of different departments.

- Software Development (/24) supports 254 usable host addresses.
- Quality Assurance (/25) supports 126 usable host addresses.
- Human Resources (/26) supports 62 usable host addresses.
- Executive Management (/27) supports 30 usable host addresses.

The remaining IP address range (10.10.1.224 – 10.10.255.255) is available for future expansion. This subnetting strategy ensures efficient IP address utilization, improved security, better network performance, and simplified network management.

QUESTION 3: IPv6 Network Configuration for Smart Hospital

Given:

- IPv6 Network Address = 2001:0db8:abcd:0000:0000:0000:0000/64
- Prefix Length = /64

Step 1: Compress the IPv6 Address

Original IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Compression Rules:

- Remove leading zeros in each block.
- Replace consecutive groups of all zeros with "::" (only once).

Compressed IPv6 Address

2001:db8:abcd::/64

Step 2: Expand the IPv6 Address

Compressed IPv6 Address

2001:db8:abcd::/64

Expanded IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Determine the Network Prefix

Prefix Length = /64

The first 64 bits represent the Network Prefix.

Network Prefix

2001:db8:abcd:0::/64

Step 4: Calculate the Total Number of Host Addresses

Host Bits

= $128 - 64$

= 64

Total Possible Host Addresses

= 2^{64}

= 18,446,744,073,709,551,616

Step 5: Total Number of Possible Host Addresses

IPv6 provides

2^{64}

possible host addresses within a /64 network.

Therefore,

Total Host Addresses

= 18,446,744,073,709,551,616

Final Result Table

Description	Result
Original IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Compressed IPv6 Address	2001:db8:abcd::/64
Expanded IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Prefix Length	/64
Network Prefix	2001:db8:abcd:0::/64
Host Bits	64

Description	Result
Total Possible Host Addresses	18,446,744,073,709,551,616 (2^{64})

Interpretation of Results

The hospital's IPv6 network address is successfully compressed by removing leading zeros and replacing consecutive zero blocks with "::". The expanded address restores the complete hexadecimal representation.

Using the /64 prefix, the first 64 bits identify the network and the remaining 64 bits are reserved for host addresses. This allows 2^{64} (18,446,744,073,709,551,616) possible host addresses, making IPv6 ideal for supporting thousands of connected medical devices, IoT sensors, and future network expansion in a smart hospital.

QUESTION 4: Routing Table Analysis

Given:

- Routing Table Entries:
 - 192.168.1.0/24
 - 192.168.2.0/24
 - 192.168.3.0/24
- Destination IP Address = 192.168.2.45

Step 1: Determine the Destination Subnet

Subnet Mask = /24
 Subnet Mask in Decimal = 255.255.255.0

Network Address

192.168.2.45 AND 255.255.255.0

= 192.168.2.0

Therefore,

Destination Subnet

= 192.168.2.0/24

Step 2: Find the Matching Routing Table Entry

Routing Table

192.168.1.0/24

192.168.2.0/24 ← Match

192.168.3.0/24

Matching Entry

= 192.168.2.0/24

Step 3: Determine the Next-Hop Network / Interface

Since the destination belongs to subnet
192.168.2.0/24,

the router forwards the packet through the interface or next-hop configured for the
192.168.2.0/24 network.

Next-Hop Network

= 192.168.2.0/24

Outgoing Interface

= Interface connected to the 192.168.2.0/24 network.

Step 4: Determine the Default Route=

If the destination IP address does not match any routing table entry,

the router forwards the packet using the Default Route.

Default Route

= 0.0.0.0/0

This route forwards packets to the default gateway for unknown destinations.

Final Result Table

Description	Result
Destination IP Address	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Routing Table Entry	192.168.2.0/24
Next-Hop Network	192.168.2.0/24
Outgoing Interface	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0

Interpretation of Results

The destination IP address 192.168.2.45 belongs to the subnet 192.168.2.0/24. Therefore, the router matches the routing table entry 192.168.2.0/24 and forwards the packet through the interface connected to that network.

If no matching route exists, the router uses the default route (0.0.0.0/0) to forward the packet to the default gateway. This routing process ensures that packets reach their correct destination efficiently and supports reliable communication across interconnected networks.

QUESTION 5: LAN Configuration Using a Layer-3 Router

Given:

- Administration LAN = 192.168.10.0/26
- Finance LAN = 192.168.10.64/26

Router Interfaces:

- Administration Gateway = 192.168.10.1
- Finance Gateway = 192.168.10.65

Step 1: Administration LAN (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2 \\ &= 62 \text{ Hosts}\end{aligned}$$

Network Address : 192.168.10.0
First Host Address : 192.168.10.1
Last Host Address : 192.168.10.62
Broadcast Address : 192.168.10.63

Suggested IP Addresses for Computers

Computer 1 : 192.168.10.2
Computer 2 : 192.168.10.3
Computer 3 : 192.168.10.4

Default Gateway

192.168.10.1

Step 2: Finance LAN (/26)

Subnet Mask = 255.255.255.192

Formula:

$$\begin{aligned}\text{Usable Hosts} &= 2^{(32 - 26)} - 2 \\ &= 64 - 2 \\ &= 62 \text{ Hosts}\end{aligned}$$

Network Address : 192.168.10.64
First Host Address : 192.168.10.65
Last Host Address : 192.168.10.126
Broadcast Address : 192.168.10.127

Suggested IP Addresses for Computers

Computer 1 : 192.168.10.66
Computer 2 : 192.168.10.67
Computer 3 : 192.168.10.68

Default Gateway

192.168.10.65

Final Result Table

Description	Administration LAN	Finance LAN
Network Address	192.168.10.0	192.168.10.64
Subnet Mask	/26 (255.255.255.192)	/26 (255.255.255.192)
Valid Host Range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Broadcast Address	192.168.10.63	192.168.10.127
Usable Hosts	62	62
Computer 1 IP Address	192.168.10.2	192.168.10.66
Computer 2 IP Address	192.168.10.3	192.168.10.67
Computer 3 IP Address	192.168.10.4	192.168.10.68
Default Gateway	192.168.10.1	192.168.10.65

Interpretation of Results

The company network is divided into two separate /26 LANs using a Layer-3 router.

- Administration LAN has the network address 192.168.10.0 with 62 usable host addresses.
- Finance LAN has the network address 192.168.10.64 with 62 usable host addresses.

The router interfaces (192.168.10.1 and 192.168.10.65) act as the default gateways for their respective LANs. All hosts in each LAN should use the corresponding gateway to communicate with devices in other networks. This subnetting design improves network organization, security, efficient IP address utilization, and reliable communication between departments.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

